

GEETHIKA KARANAM

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EDUCATION

North Carolina State University

Master of Science in Engineering/Industrial Management

Raleigh, NC

May 2025

Mahindra University, Ecole Centrale College of Engineering

Bachelor of Technology in Electrical Engineering

Hyderabad, India

June 2023

SKILLS

Programming Languages: Python, R, SQL | **ML & Data Science libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV

Analytics and Data Visualization tools: Excel (Solver, VBA), JMP, Tableau, PowerBI, Google Analytics, Looker, Statistical Analysis (ANOVA, Chi-Squared testing)

Database Management tools: PostgreSQL, MySQL, MongoDB, Snowflake, MS Access

Project Management & workflow tools: Jira, Smartsheet, Notion, MS Project

PROFESSIONAL EXPERIENCE

Trinity Health

Raleigh, NC

Supply chain Analyst Intern (Capstone Course, NCSU)

January 2025 – May 2025

- **Led the requirements gathering** for the redesign of Trinity Health's supply chain data management system, **analyzing 100K+ historical data** on bills of materials (BOMs) in **python** to identify insights critical to improving demand planning, payment accuracy, and costing precision.
- **Collaborated with key stakeholders**, including Database administrators, SCM procurement leaders, to uncover errors in the current SSID (Strategic Sourcing Implant Database) and thus implement a new seamless strategy for Epic EMR using standardized SOPs, by a comparative databases analysis.
- **Created interactive data visualizations using PowerBI** to communicate supply chain insights like cost trends, part usage frequency, and procedural variation to cross-functional stakeholders, highlighting critical data quality issues and opportunities for enhanced data governance.
- **Compiled a comprehensive report** evaluating Trinity Health's transition from the legacy SSID procurement system to a modernized framework. Conducted data cleaning, comparative analysis, and visualization to identify data quality gaps and standardization opportunities, ultimately providing actionable insights that support improved implementation strategies for the new system.

Skills- Data Analysis, Supply Chain Analysis, Supply Chain Management, Project Management

Data Science and AI Academy at North Carolina State University

Raleigh, NC

Student Program Assistant

January 2025 – May 2025

- Assisting in hosting workshops, conferences and datathons through the academy (**Women in Data Science @ Raleigh and the ASA Datafest 2025**) to foster the inter-disciplinary data science community at NC State University.
- Analyze participant data and feedback for program evaluations, using Excel and PowerBI to generate reports that inform future event planning strategies and marketing ideas.
- Engage with and mentor students in the Data Science and AI Academy, providing academic and career guidance to facilitate their success in data science education.

Skills- Organizational skills, Time management, Interpersonal communication

Schneider Electric

Hyderabad, India

Data Analyst Intern

July 2022 - August 2022

- **Designed and implemented a robust data validation framework** to detect and mitigate sensor malfunctions, ensuring real-time accuracy and **reducing variance by 15.2%** through improved calibration and wiring integrity.
- **Optimized real-time sensor data transmission** using Zigbee-based IoT systems, integrating a cloud-based data management system with preprocessing techniques to **enhance data integrity and operational efficiency**.
- **Developed an interactive real-time dashboard for shop floor monitoring, visualizing key metrics like humidity, temperature, and pressure**, empowering management with actionable insights for **proactive decision-making**.

Skills- Sensor data analysis, Python scripting, IOT Protocols

COMPETITIONS

Women in Data Science Datathon 2025

- Built a predictive Support Vector Classifier to determine participant sex from fMRI (Functional MRI) data, using **ANOVA for feature selection** and **PCA for dimensionality reduction**, retaining 90% variance across 10,000+ features.
- **Optimized model performance through grid search with cross-validation**, strategically selecting hyperparameters based on F1 score to handle class imbalance in the dataset.
- **Achieved 83% accuracy and 80% macro F1 score**, demonstrating the model's effectiveness predicting sex, later integrated with a model for predicting patient ADHD Outcome.

Skills-Python, Supervised and Unsupervised ML, Data Pre-processing Model Evaluation (Precision, Recall, F1), Hyperparameter Tuning

PROJECTS

Wolfpack Outfitters sales and purchase Analysis | [github](#)

- **Initiated and led the project requirements gathering with key stakeholders at Wolfpack Outfitters (the official merchandising store of NCSU)** to identify the **NorthStar KPIs** for analyzing the sales trends and purchase patterns for the product 315- Hooded Sweatshirt, ensuring alignment with business goals and objectives.
- **Designed a comprehensive use-case diagram** using **Visual Paradigm** to map data flow and user interactions, fostering stakeholder engagement and ensuring continuous alignment with project scope and client requirements.
- Conducted extensive **exploratory data analysis (EDA)** on multi-channel sales data, leveraging **MySQL** to clean, validate, and transform datasets for brick-and-mortar and online sales, ensuring high data integrity and accuracy.
- **Developed a dynamic Power BI dashboard** to visualize key sales KPIs, month-over-month purchase trends, and seasonal demand fluctuations and forecasting demand for the next fiscal year providing actionable insights into customer buying behavior.
- **Presented strategic recommendations to key stakeholders**, including optimizing online sales through website diagnostics analysis and implementing a data-driven inventory management strategy to enhance stock replenishment and promotional opportunities for least popular listings.

Skills-Visual Paradigm, SQL, PowerBI, Stakeholder management

LEADERSHIP/VOLUNTEERING

- **WiDS (Women in Data Science) Ambassador at North Carolina State University**
- **Team Lead, WiDS Datathon 2025:** Leading a team in developing machine learning models for brain activity pattern recognition to enhance ADHD detection, leveraging machine learning.
- **Market Data Analyst Intern at Seaside Sustainability (NPO):** Collaborated with a cross-function team to conduct a market size using Google Trends, analyzing trends and customer behavior. Reporting insights to stakeholders helping device sponsorship strategies aligning with Seaside's mission of ocean and beach cleaning initiative.
- **DataKind Volunteer:** Contributed to the development of the Data Observation Toolkit (DOT), an open-source solution for improving data quality in digital health systems, now recognized as digital public good